How to read the FlexiCT error-review outputs

Final-block fine-tuning | CV_1 | Five-fold ensemble | 100% development data

These outputs address three different questions: **What went wrong? Which regions receive attribution? Does changing those regions affect the prediction?** They do not, by themselves, establish why an anatomical diagnosis was missed. The model is final-block fine-tuning, CV_1, five-fold probability ensemble, using 100% of development data. It made 20 errors among 278 test cases.

1. Error inventory and class-confusion table

`misclassified_cases.csv` lists all errors, with the most confident first.

Output	Meaning	Useful interpretation
`true_class`	Recorded dataset label	The comparison label; it is not independently adjudicated by this notebook.
`predicted_class`	Class with highest mean probability across the five folds	What the evaluated ensemble actually selected.
`predicted_probability`	Ensemble probability assigned to its selected class	High values identify confident errors; 0.99 does not mean a calibrated 99% chance of correctness.
`true_probability`	Ensemble probability assigned to the recorded class	A low value means the ensemble gave the recorded label little support.
`fold_votes_for_ensemble_prediction`	Number of individual fold models whose top class equals the ensemble class	5/5 means consistent prediction across folds. It can still be consistently wrong.
`replay_max_abs_difference`	Largest difference between saved and reproduced class probabilities	Technical reproducibility of the explanation run. It does not validate attribution quality.

The ensemble averages probabilities, not votes. It is therefore possible for its winning class to receive only 2/5 fold votes, as happens in this review. The five models have overlapping development training sets, so their votes are not five independent diagnostic opinions.

`error_pairs.csv` and the notebook crosstab contain **errors only**. There are 9 recorded disease cases predicted as Normal, 8 recorded Normal cases predicted as a disease class, and 3 confusions between disease classes. These counts do not provide class-specific error rates without the total test count per class.

2. CT overview: `*_ct_overview.png`

Twelve evenly spaced axial slices show the image volume actually supplied to the model after resampling and center cropping/padding. Inspect coverage, visibility, and image quality before interpreting the heatmap. This overview is not the full original-resolution examination, and twelve selected slices can miss a small finding. Use the optional slice browser to inspect all 160 model slices and the original NIfTI for a complete clinical review.

3. Attribution image: `*_attribution.png`

Rows are axial, coronal, and sagittal planes through the maximum of the predicted-class attribution. These planes are chosen by the attribution, not by an annotated lesion.

Column	What it shows	What it does not show
CT	The model-space anatomy without an overlay	The entire original scan
Predicted-class attribution	Positive Grad-CAM attribution for the class the ensemble chose; brighter regions have larger displayed attribution	A verified causal explanation or lesion mask
Recorded-class attribution	Positive attribution for the recorded label, even when its probability was low	A map of where the disease must be
Contrast	Raw predicted-class CAM minus raw recorded-class CAM; red is higher for the predicted target, blue higher for the recorded target	A probability map, diagnostic boundary, or simple healthy/diseased distinction

The two positive heatmaps each have their own 0-1 display scale. A bright recorded-class map can coexist with a very low recorded-class probability. Brightness cannot be compared numerically between the two columns or patients. The contrast is computed before those independent display scalings.

The native attribution grid has 16 mm spacing. Smooth interpolation does not turn it into a precise anatomical boundary. A heatmap for the Normal class also does not identify a specific normal organ or prove absence of disease.

4. Occlusion table and `occlusion_diagnostic.png`

Occlusion tests the map by replacing regions of the model input and evaluating the unchanged ensemble again. The six highest-ranked 40-voxel cubes out of 64 are masked together: 9.375% of the input. At 2 mm spacing, each cube has an 80 mm edge, so these are coarse perturbations.

The reported quantity is:

```
probability drop = original predicted-class probability - masked probability
```

- **Positive:** masking decreased that probability.
- **Zero:** no measured change.
- **Negative:** masking increased that probability.

Column	Meaning
`top_mask_probability_drop`	Drop after masking the highlighted cubes with normalized mean intensity
`random_mask_mean_probability_drop`	Average drop for three randomly placed masks with the same number of cubes
`top_minus_random_drop`	Top-mask drop minus random-mask mean drop; measures their relative difference
`air_fill_probability_drop`	Drop for the same highlighted cubes with an air-like replacement intensity
`top_drop_exceeds_random`	Whether the top-mask drop exceeds the random mean, regardless of sign

Column	Meaning
`top_mask_reduces_probability`	Whether the top-mask drop is actually positive
`positive_drop_exceeds_random`	Both of the preceding conditions; descriptive positive-evidence support under this perturbation

The sign must be checked before interpreting the comparison. In this run, all 20 top-mask drops exceed their random controls, but only 14 are positive. The other six increase predicted-class probability. This corrects the earlier overly broad interpretation that all 20 cases supported positive evidence. Even a positive difference can be tiny, and three random masks are not enough for a formal significance claim.

Two worked examples:

- **Normal_R01:** predicted Appendicitis with probability 0.9994 and 5/5 fold votes. Highlighted-region masking drops that probability by 0.9873 (98.73 percentage points), versus 0.0303 (3.03 points) for random masking. These regions matter strongly under this test despite the recorded prediction error. Air-like masking gives a smaller 0.3399 drop, showing that the replacement intensity changes the effect size. This does not establish appendicitis or prove the recorded label wrong.
- **Appendicitis_R17:** predicted Normal with probability 0.5901 and 3/5 fold votes. Highlighted-region masking has a drop of -0.0851: the probability actually rises to about 0.6751. Random masks increase it more, producing a mean drop of -0.1236. The positive top-minus-random difference is therefore insufficient evidence that the highlighted regions positively support Normal. Air-like masking gives a small positive drop of 0.0155, further indicating sensitivity to the chosen masking intervention.

Masking creates images the model may not have encountered during training. The checks therefore assess response to a specified intervention, not a universal causal explanation of diagnosis.

5. Fold consistency and randomized-head diagnostics

`fold_probabilities.csv` contains each model's five probabilities. It helps separate unanimous errors from disagreements concealed by the ensemble average.

`mean_pairwise_fold_map_spearman` compares the spatial rankings in the five predicted-class positive maps. Values near 1 indicate similar rankings; near 0 little rank association; negative values indicate opposing rankings. Repeatable maps can still highlight inappropriate features. There is no validated good/bad cutoff in this notebook.

`randomized_head_map_spearman` compares fold 1's original map with its map after randomizing only the final classifier layer. A large change suggests the map depends on learned classifier weights. Similar maps would raise questions about that dependence. Low or negative correlation alone does not validate an explanation, and this is only a limited head check rather than the complete published randomization protocol.

6. Coverage and background diagnostics

`crop_removed_xyz_mm` reports the total resampled extent removed along each axis by center cropping. For example, `52/52/118` means 52 mm removed along X, 52 mm along Y, and 118 mm along Z, counting both ends of each axis. It does not say whether a lesion was removed; check the original scan and crop.

`attribution_fraction_below_minus900hu` is the fraction of positive predicted-class attribution lying in voxels below -900 HU. A high value is a reason to inspect air, lung, bowel gas, padding, and image borders. The threshold does not distinguish these structures, and a high value alone does not prove the use of an irrelevant shortcut.

7. Expert review and reproducibility files

`radiologist_review.csv` is the human review worksheet. The key questions are whether the relevant finding is in the model crop, whether the highlighted location is anatomically meaningful, and whether image quality or label ambiguity needs investigation. Blank fields represent review still to do.

The `.npz` files retain the actual input and attribution arrays, while `preprocessing.json`, `summary.json`, execution JSONs, and the project log record how outputs were produced and checked. Hashes detect changed inputs or artifacts; they are not measures of scientific accuracy.

Start with the CT overview, inspect both class maps, then check the sign and size of the occlusion effects, fold consistency, and crop coverage. Record an anatomical conclusion only after reviewing the actual findings.

The publication references and method adaptations are listed in README.md: Grad-CAM (Selvaraju et al., 2017), occlusion analysis (Zeiler and Fergus, 2014), saliency sanity checks (Adebayo et al., 2018), and the 2021 Radiology: Artificial Intelligence study of saliency reliability.

Publication references

1. Selvaraju RR et al. Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization. ICCV, 2017. https://arxiv.org/abs/1610.02391

2. Zeiler MD, Fergus R. Visualizing and Understanding Convolutional Networks. ECCV, 2014. https://arxiv.org/abs/1311.2901

3. Adebayo J et al. Sanity Checks for Saliency Maps. NeurIPS, 2018. https://arxiv.org/abs/1810.03292

4. Assessing the Trustworthiness of Saliency Maps for Localizing Abnormalities in Medical Imaging. Radiology: Artificial Intelligence, 2021;3(6). https://doi.org/10.1148/ryai.2021200267